

Problem set 5

Clustering

Excercise 1

Compute the confusion matrix, macro-averaged Precision, Recall, and F-score for the clustering shown below.

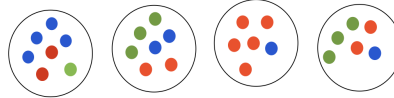


Figure 1: Caption

Excercise 2

For the same clusters as in Excercise 1, compute B-CUBED Precision, Recall, and F-score.

Excercise 3

Given the following set of 2-dimnesional points $X_1 = (2, 10)$, $X_2 = (2, 5)$, $X_3 = (8, 4)$, $X_4 = (5, 8)$, $X_5 = (7, 5)$, $X_6 = (6, 4)$, $X_7 = (1, 2)$, $X_8 = (4, 9)$ apply single-linkage agglomerative clustering using Euclidean distance. Draw a dendrogram representing the resulting hierarchy of clusters.

Excercise 4

The dataset consists of four objects with pairwise distances given by the following distance matrix

$$\begin{pmatrix} & X_1 & X_2 & X_3 & X_4 \\ X_1 & 0 & 2 & 6 & 2.5 \\ X_2 & 2 & 0 & 3 & 8 \\ X_3 & 6 & 3 & 0 & 4 \\ X_4 & 2.5 & 8 & 4 & 0 \end{pmatrix}.$$

1. Cluster dataset using single-linkage agglomerative clustering. Draw the corresponding dendrogram.

2. Cluster dataset using complete-linkage agglomerative clustering. Draw the corresponding dendrogram.